

C.1 Joint Distributions

L1 Joint Distributions

- In 1 dim., for RV X we had $\sum P_X(X=x) = 1$
- In 2 dim., we have 2 RV X & Y .

Joint PMF is $P_{XY}(X=x, Y=y)$

Total Probab. $\Rightarrow \sum_x \sum_y P_{XY}(X=x, Y=y) = 1$

→ currently we deal with only discrete joint distributions.

Ex.

	X		
	0	1	2
Y	0.1	0.2	0.3
2	0.1	0.1	0.2

$$P_{XY}(X=0, Y=1) = 0.1$$

$$P_{XY}(X=2, Y=2) = 0.2$$

$$\sum = 1$$

Ex. $1 \leq X \leq 4$, $1 \leq Y \leq X$. All are equally likely, $P(X=Y=3)$

$$(X, Y) \Rightarrow (1, 1)$$

$$(2, 1), (2, 2)$$

$$(3, 1), (3, 2), (3, 3)$$

$$(4, 1), (4, 2), (4, 3), (4, 4)$$

$$\Rightarrow \frac{1}{10}$$

$$10 \Rightarrow \frac{1}{10}$$

- In 1d, the CDF of X is $F_X(x) = P(X \leq x)$
 In 2d, the joint CDF of X & Y is

$$F_{X,Y}(x,y) = P(X \leq x, Y \leq y)$$

Properties of joint PDF is -

1. $0 \leq F(x,y) \leq 1$
2. $F(x, \infty) = P(X \leq x) = F_X(x)$
3. $F(\infty, y) = P(Y \leq y) = F_Y(y)$
4. $F(\infty, \infty) = 1$
5. $F(-\infty, y) = F(x, -\infty) = 0$

L2 Marginal & Conditional Distributions

- For discrete variables.

$$P(X=x) = \sum_y P(X=x, Y=y)$$

↳ is the marginal distribuⁿ of X .

Also, we can say that it's the distribuⁿ of X without knowing Y .

eg. $p(x,y) = \begin{pmatrix} 2x+y \\ 1, 2 \end{pmatrix}$

for $(x,y) = (0,1), (0,2), (1,2), (1,3)$
 & 0 otherwise.

Find marginal PF for X .

$$P(X=0) = P(X=0, Y=1) + P(X=0, Y=2) \\ = \frac{1}{12} + \frac{2}{12} = \frac{3}{12}$$

$$P(X=1) = P(X=1, Y=2) + P(X=1, Y=3) \\ = \frac{4}{12} + \frac{5}{12} = \frac{9}{12}$$

Eg. same eq. $P(X=0 | Y=2)$

$$P(X=0 | Y=2) = \frac{P(X=0, Y=2)}{P(Y=2)}$$

$$P(Y=2) = P(X=0, Y=2) + P(X=1, Y=2) \\ = \frac{2}{12} + \frac{4}{12} = \frac{6}{12}$$

$$\frac{P(X=0, Y=2)}{P(Y=2)} = \frac{2/12}{6/12} = \frac{1}{3}$$

- For discrete vars.

$$P(X=x | Y=y) = \frac{P(X=x, Y=y)}{P(Y=y)}$$

$$= \frac{P(X=x, Y=y)}{\sum_x P(X=x, Y=y)}$$

- The X & Y are said to be independent if $P(X=x | Y=y) = P(X=x)$

$$P(X=x, Y=y) = P(X=x) \cdot P(Y=y)$$

if X & Y are independent.

($\rightarrow$) X & Y are independent iff. the joint PMF factors as a funcⁿ of x times a funcⁿ of y & the range of $\{X, Y\}$ is a discrete rectangle.

- There are few key points to remember -
1. The marginal distribuⁿ $P(X=x)$ of X can only involve x . It does not involve any other variable.
 2. Conditional distribuⁿ $P(X=x | Y=y)$ can involve x & y .
 3. If X & Y are independent,

$$P(X=x | Y=y) = P(X=x) \text{..}$$

The conditional distribuⁿ of X will only involve x . Even the range can't depend on y .